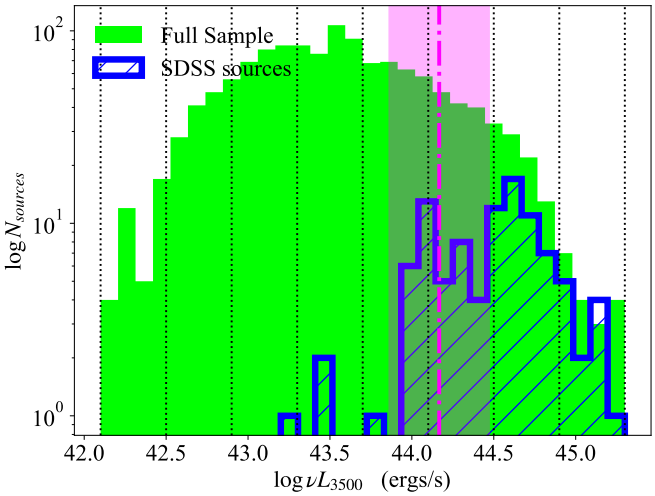

Figure 4. The 3500 Å luminosity distribution for the sources within the $\log_{10} M/M_\odot \in [8.0, 8.5]$ mass bin. The green bars show the distribution for the full sample, while the blue-dashed bars show the distribution of the sources in the SDSS DR16 AGN catalogue (Wu & Shen 2022). The shaded magenta region band shows the 3500 Å monochromatic luminosity at 2 % of Eddington for the black hole mass range $\log_{10} M/M_\odot \in [8.0, 8.5]$, calculated for a standard disc extending to the ISCO. The dashed-dotted magenta line shows this for the geometric centre of the mass-bin. The vertical black dotted lines show the luminosity binning used in the SED analysis in this paper. It is clear that not only does our X-ray selected sample extend to considerably lower optical luminosity than the SDSS matched sample, but it also peaks below 2 % of Eddington, unlike the SDSS matched sample.

3 MODELLING THE SEDS

To explore the spectral evolution in detail we fit the stacked AGN SEDs within our fiducial mass bin of $\log_{10} M/M_\odot \in [8.0, 8.5]$. We do not use standard Shakura & Sunyaev (1973) disc models, as these have multiple well known issues. Firstly, there is the high energy X-ray tail which is always present at a significant level (Elvis et al. 1994; Lusso & Risaliti 2016). More seriously for the disc models themselves, the UV spectrum often shows a downturn before the predicted peak, which appears to connect across the data-gap (due to ISM absorption, e.g Laor et al. 1997; Telfer et al. 2002) to an upturn in the soft X-ray bandpass (the soft X-ray excess) (e.g Czerny & Elvis 1987; Laor et al. 1997; Mehdipour et al. 2011; Jin et al. 2012; Petrucci et al. 2018; Kubota & Done 2018). The standard disc models also predict that the time-scales for intrinsic variability is extremely slow (thousands of years), in conflict with multiple observations (e.g Noda & Done 2018; Stern et al. 2018; Hernández Santisteban et al. 2020; Neustadt & Kochanek 2022; Yao et al. 2023; Neustadt et al. 2024). Nonetheless, the ubiquity of the strong UV emission clearly indicates that there is optically thick material emitting a large fraction of the gravitational potential energy from accretion, i.e there is a disc but its properties (spectrum and variability) are different to those predicted by standard models.

Hence we use modified disc models, which assume that the disc vertical structure can be different than in standard models such that it does not completely thermalise (Różańska et al. 2015; Jiang & Blaes 2020; Kawanaka & Mineshige 2024). This could then form a warm Comptonising plasma, which is optically thick, giving rise to a Comptonised disc-spectrum, rather than a pure multi-colour black-body disc spectrum; naturally explaining the UV turnover linking into the soft X-ray excess (Mehdipour et al. 2011; Done et al. 2012; Mehdipour et al. 2015; Kubota & Done 2018; Petrucci et al. 2018), and generally comprises the dominant component of the disc emission (Czerny & Elvis 1987; Magdziarz et al. 1998; Done et al. 2012; Petrucci et al. 2013, 2018). However, the model does still allow the emission to revert back to a standard black-body disc at larger radii (Done et al. 2012; Jin et al. 2012; Kubota & Done 2018).

In addition to a disc and disc-like structure, the model allows for the accretion flow to evaporate into an optically thin hot X-ray plasma in the innermost regions, $r_{\rm isco} < r < r_{\rm hot}$, giving rise to the Comptonised high energy X-ray tail.

Hence, our accretion structure consist of three distinct regions: An inner X-ray plasma for $r_{\rm isco} < r < r_{\rm hot}$, which then condenses into a warm optically thick disc-like structure for $r_{\rm hot} < r < r_{\rm warm}$, and then eventually reaching thermalisation forming a standard disc for $r > r_{\rm warm}$ (Kubota & Done 2018). These models generally give a good fit to the observed SEDs of individual AGN with significantly higher quality/broader bandpass data (e.g Matt et al. 2014; Matzeu et al. 2016; Done & Jin 2016; Czerny et al. 2016; Hagino et al. 2017; Porquet et al. 2018, 2019, 2024), and comes with the additional feature that the non-standard optically thick region gives the potential for significantly faster intrinsic disc variability (Hagen et al. 2024), more in line with current observations.

In this section we give a brief overview of the physical implementation of the model in section 3.1, using the publicly available XSPEC model AGNSED (Kubota & Done 2018), before applying it to the data in 3.2. Throughout we use the standard notation for radii, where R denotes the radius in physical units and r is the dimensionless radius in gravitational units, related through $R = rR_G$ where $R_G = GM/c^2$.

3.1 AGNSED - Physical Implementation

AGNSED (see Kubota & Done 2018 for full details) assumes a standard (relativistic) disc Novikov & Thorne (1973) emissivity throughout the entire flow, with a radial temperature profile given by $T^4_{NT}(R) \propto R^{-3} f(R)$ where $f(R)$ describes the radial disc structure in the Kerr metric (Page & Thorne 1974).

In the outermost regions of the flow, $r > r_{\rm warm}$, this model assumes the flow thermalises to a standard accretion disc. Hence, splitting the flow up into radial annuli of width ΔR, the emission from each annulus will be a standard black-body $B_\nu(T_{NT}(R))$ with total luminosity $2 \times 2\pi R \Delta R \sigma T^4_{NT}(R)$, where the extra factor 2 comes from the disc emitting from both sides and σ is the Stefan-Boltzmann constant.

In the warm disc-like region, $r_{\rm hot} < r < r_{\rm warm}$, the vertical disc structure is assumed to change, such that the dissipating region moves up into the photoshpere forming a slab geometry above an underlying passive disc structure. Here the seed photons are tied to the underlying disc, such that their temperature is simply T_{NT} and each annulus has luminosity $4\pi R \Delta R \sigma T^4_{NT}(R)$. Additionally, the seed photons are assumed to form a black-body distribution, which then Compton scatters through the photosphere forming a Comptonised black-body spectral shape; modelled internally using NTHCOMP (Zdziarski et al. 1996; Zycki et al. 1999). It is assumed that the disc photosphere is optically thick, and has a covering fraction of unity such that all seed-photons are Comptonised (Petrucci et al. 2018). Within AGNSED the warm Comptonisation region is assumed to have a single electron temperature, $kT_{e,w}$, and photon index, Γ_w, left as free parameters.

Below a radius $r_{\rm hot}$ the warm disc-like structure evaporates into a hot, optically thin, geometrically thick X-ray plasma; responsible for forming the high energy X-ray tail (Narayan & Yi 1995; Liu et al. 1999; Różańska & Czerny 2000). This plasma is powered by the energy dissipated in the flow between $r_{\rm hot}$ and $r_{\rm isco}$, still assum-

$\log \nu L_{3500}$ (ergs/s)	$\log \dot{m}$	$kT_{e,w}$ (keV)	Γ_h	Γ_w	$r_{\rm hot}$	T_{hi} (K)	$[r_w(T_{hi})]$
[42.1, 42.5]	−1.93	†	1.53	†	306††	4611	[306]
[42.5, 42.9]	−2.04	†	1.65	†	224††	5464	[224]
[42.9, 43.3]	−2.04	†	1.79	†	189††	6054	[189]
[43.3, 43.7]	−1.93	0.36	1.92	2.60	115	5920	[217]
[43.7, 44.1]	−1.80	0.42	1.93	2.72	52.8	5920*	[238]
[44.1, 44.5]	−1.40	0.46	1.69	2.63	23.0	5920*	[328]
[44.5, 44.9]	−0.98	0.20	1.71	3.82	15.8	5920*	[463]
[44.9, 45.3]	−0.61	0.99	2.35	3.70	12.4	5920*	[610]

Table 1. Fit values for each of the luminosity bins (left column) for the stacked SEDs within the mass bin $\log_{10} M/M_\odot \in [8.0, 8.5]$. We only show parameters that were kept free during the fitting process. The r_w values in square brackets show the radius where the disc temperature is T_{hi}. Note that the errors on the derived parameters are too large to be meaningful (due to the significant dispersion in the stacks).
† The warm Compton region is not required in this bin.
†† No warm Compton region is required so $r_w = r_h$, and $T_{hi} = T_{bb}(r_h)$.
∗ In these bins the standard outer disc is subdominant and T_{hi} cannot be well constrained. Hence we freeze it to the last bin where it could make an impact (i.e [43.3, 43.7]).

ing standard Novikov & Thorne (1973) emissivity, giving $L_{\rm diss}$. In addition there is a contribution to the total emission from the seed-photons entering the corona from the disc, $L_{\rm seed}$, such that the total power emitted by the corona is $L_{\rm hot} = L_{\rm diss} + L_{\rm seed}$ (see Kubota & Done 2018 for details on calculating thse). As with the warm Comptonisation in the disc-like region, it is assumed that the X-ray plasma has a single electron temperature $kT_{e,h}$ and photon index Γ_h, which are left as free parameters.

The total SED is then simply found by adding together the integrated emission from each component within the flow.

A key feature of AGNSED is the energy balance. The total available power within the flow is always set by the mass, M_{BH}, and Mass-Accretion rate, $\dot{M}$, assumed constant at all radii in the flow. The relative contribution to the total luminosity from each of the three regions within the flow is then set by the integrated emissivity over the radii where each component dominates. As an example, a flow where the disc-like structure covers the majority of the available emitting area down to $r \sim r_{\rm isco}$ will have an SED dominated by the warm Comptonisation, whereas a flow with an extensive X-ray plasma with $r_{\rm hot} \gg r_{\rm isco}$ will have an SED dominated by the hot Comptonisation component with very little disc. In this sense, fitting the AGN SED gives us an estimate of not only the relative contribution to the total output power from each component, but also the relative size scales of each component.

An important note is that although AGNSED takes into account the energy balance, it does not consider the conservation of photon number although Compton scattering does. In some sense this is allowing for some uncertainty in the source of seed photons. The model only considers seed photons originating from the disc regions, which in disc dominated states will certainly be sufficient (e.g Malzac & Belmont 2009). However, in states dominated by the X-ray plasma there could be an issue with the spectrum being photon-starved when only considering seed-photons from the disc, similar to hard state BHBs (e.g Poutanen et al. 2018), and will almost certainly require additional seed photons originating within the flow through, e.g, cyclo-scynchrotron.

3.2 Fitting the data

We fit the stacked data within each bin with the AGNSED model, using the standard χ^2 minimisation routine within XSPEC v.12.13.0c (Arnaud 1996). For simplicity we treat the 1σ dispersion on the flux within each energy/wavelength bin as the error on the flux during the fitting. For mass-luminosity bins with too few sources to confidently calculate the dispersion (< 10, only relevant for the highest luminosity bin in our fiducial mass bin), we assume a 10 % error on the flux. Due to the large dispersion in the data, especially the X-ray, we stress that these fits are meant to display the rough power-output and evolution of the SED, rather than a detailed determination of model parameters.

The black hole mass is fixed to the geometric centre of the mass-bin, which for the bin considered in this paper ($\log_{10} M/M_\odot \in [8.0, 8.5]$) is $1.7 \times 10^8\, M_\odot$. We conservatively assume spin zero, where the general-relativistic corrections from ray-tracing (not included in the standard version of AGNSED) are small (see Hagen & Done 2023 - certainly smaller than the dispersion within the data). As these are unobscured AGN we assume an on average a small inclination angle, fixing it to $\cos(i) = 0.87$ throughout. Finally, we fix the X-ray coronal scale height to $h_x = 10$ (which sets the fraction of X-ray power reprocessed by the disc - see Kubota & Done (2018) for details), and the outermost radius to $r_{\rm out} = 10^4$.

Our initial fits (not tabulated) showed that the peak temperature of the outer black-body disc, $T_{hi} = T_{bb}(r_w)$ remained approximately constant ($\sim 4000 - 6500$ K), despite the large changes in $\dot{m}$. Hence, we slightly modified AGNSED such that this transition temperature is now an explicit free parameter, replacing r_w, which is now an implicit parameter calculated internally for a given $\dot{m}$ and T_{hi}.

The resulting best fit parameters are shown in Table 1, and the resulting SEDs are shown in Fig. 5. As well as the (expected) increase in $\dot{m}$ with optical luminosity, we see a correlated decrease in $r_{\rm hot}$. This shows how the inner X-ray plasma/corona becomes less and less dominant as $\dot{m}$ increases (Lusso & Risaliti 2016). This is also seen in the data themselves (Fig. 5), where the X-ray emission has comparatively little change (only a factor of a few) whereas the optical luminosity (and hence $\dot{m}$) change by multiple orders of magnitude. Our model relies on the energy balance to determine the size-scale of the components, so the approximate constant X-ray power (at

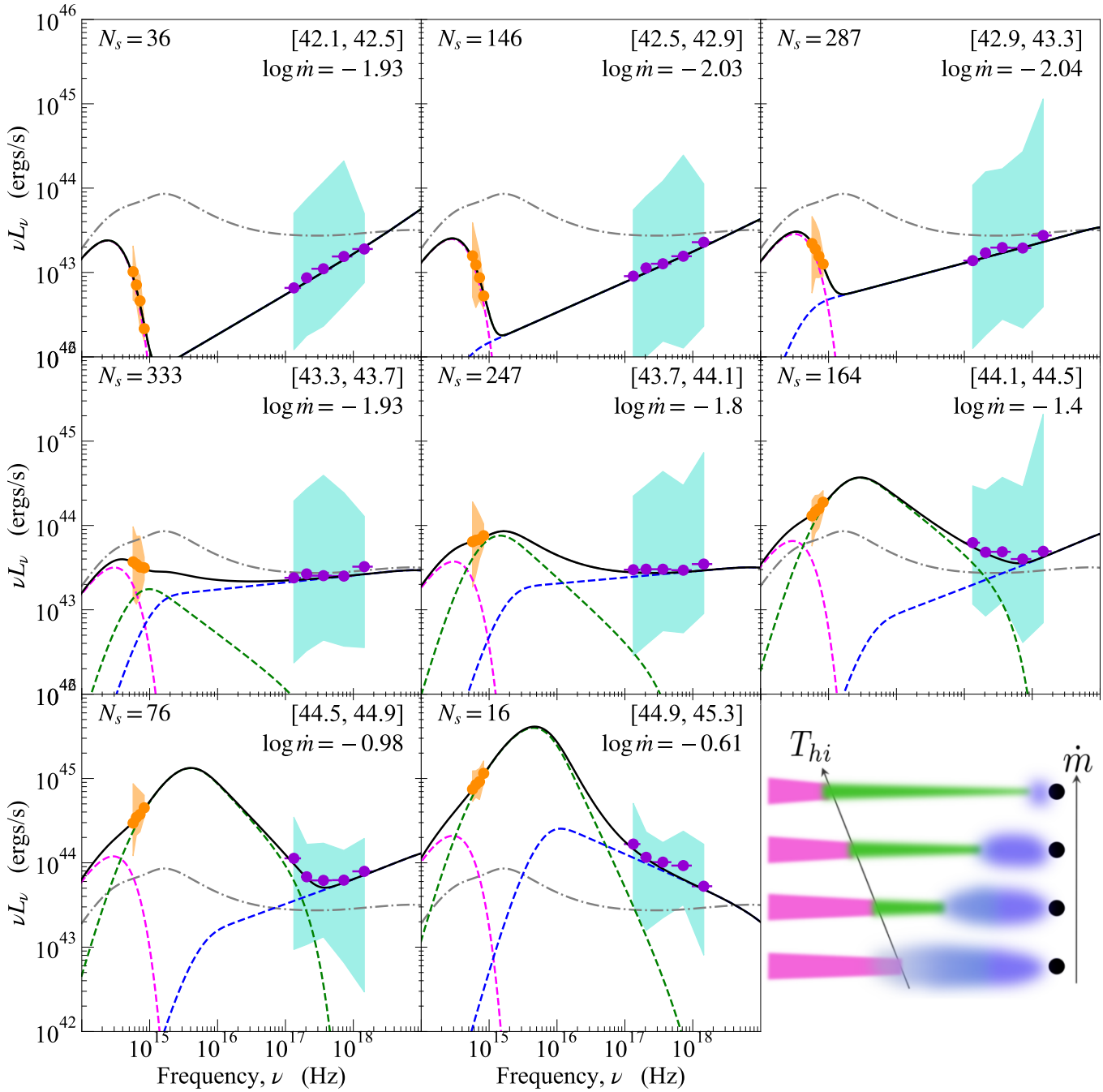

Figure 5. Stacked SEDs for the mass bin $\log_{10} M/M_\odot \in [8.0, 8.5]$. Each panel shows a single monochromatic, νL_ν, 3500 Å luminosity bin, with the precise bin range shown in ergs/s by the bracketed values in the top right corner of each panel. The number of sources, N_s, within each bin is given in the top left corner. The binned HSC data are shown by the orange points, while the binned eROSITA data are given by the violet points. The orange and cyan regions show the dispersion on the HSC and eROSITA data respectively. Each bin has been fit with AGNSED, with the resulting model SED shown by the solid black line, while the coloured dashed lines show the individual components to the total model. Here magenta indicates the outer standard disc, green the unstable warmly Comptonised disc, and blue the inner X-ray corona. The dashed-dotted grey line shows the model for the middle bin ($\log_{10} \nu L_{3500} \in [43.7, 44.1]$ ergs/s), provided to highlight the relative change in the SED as we move through the luminosity bins. The bottom right panel shows a cartoon of the change in accretion geometry with $\dot{m}$. The colours in the cartoon corresponds to the colours in the SED components.

$L_{\rm hot} \sim 0.02 L_{\rm Edd}$ - Kubota & Done 2018) means that it requires a smaller fraction of the total accretion power as $\dot{m}$ increases, and so $r_{\rm hot}$ decreases. The increasing dominance of the disc means means there are more seed photons cooling the hot corona, and so Γ_h also increases (Kubota & Done 2018). These relations have been clearly seen before (Kubota & Done 2018; Mitchell et al. 2023), but here the host galaxy subtraction also makes the warm Compton parameters more reliable. We see that there is a systematic increase in Γ_w, correlated with a decrease in $kT_{e,w}$, which implies that the warm Comptonising disc-like region is also systematically changing, from